

Vectors

An $n \times 1$ matrix

$$\begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}$$

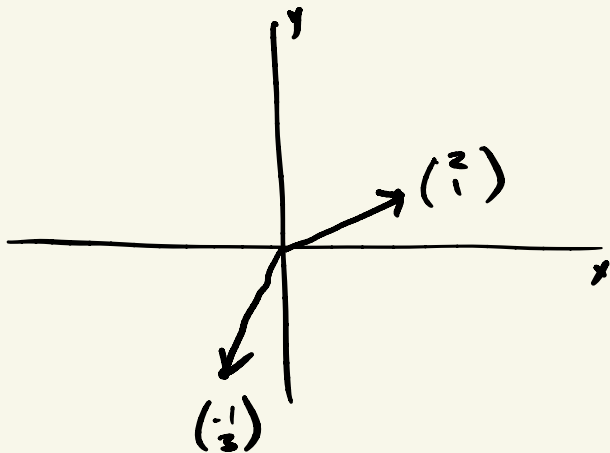
where $a_1, \dots, a_n$ are real numbers is called a (column) **vector**.

The set of all vectors with n entries is called R^n . $\rightarrow$ *n-space.*

Vectors in R^n can be added:

$$\begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} + \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix} = \begin{pmatrix} a_1 + b_1 \\ a_2 + b_2 \\ \vdots \\ a_n + b_n \end{pmatrix}$$

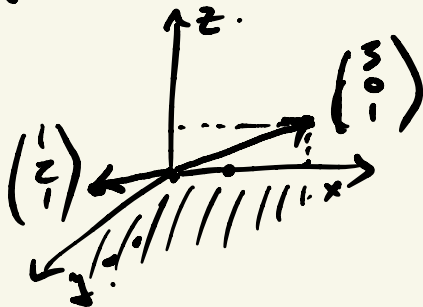
Ex: $\mathbb{R}^2$ - a plane



$$\begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

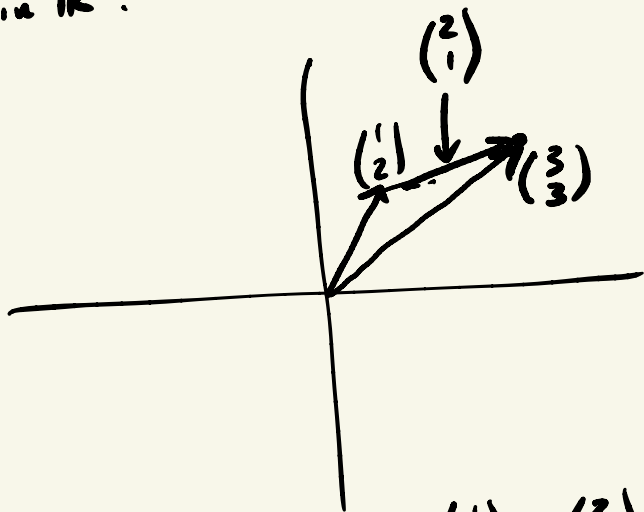
$$\begin{pmatrix} -1 \\ 3 \end{pmatrix}$$

Ex: $\mathbb{R}^3$ - 3-space usually with coords
 x, y and z .



$$\begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix}$$
$$\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$$

+ in $\mathbb{R}^2$:



$$\begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

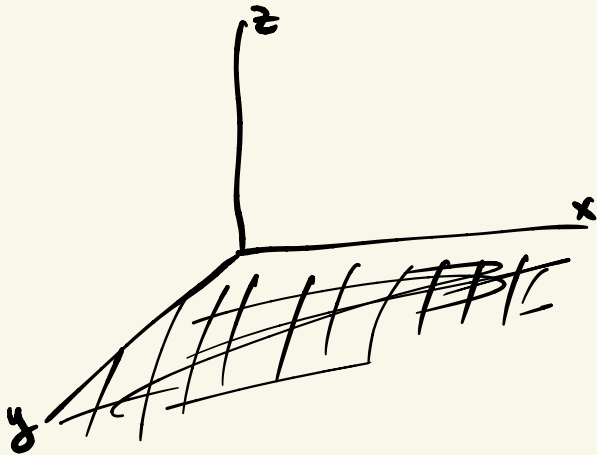
$$\begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ 2 \end{pmatrix} + \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \underline{\underline{\begin{pmatrix} 3 \\ 3 \end{pmatrix}}}$$

Ex: $1_n \mathbb{R}^3$

$$\begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix} + \begin{pmatrix} -1 \\ 2 \\ 4 \end{pmatrix} = \begin{pmatrix} \underline{0} \\ 2 \\ 2 \end{pmatrix}$$

$\mathbb{R}^2$ is not a subset of $\mathbb{R}^3$.



$$\begin{pmatrix} x \\ y \\ 0 \end{pmatrix}$$

We can also multiply vectors by scalars λ - real numbers:

$$\lambda \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} = \begin{pmatrix} \lambda a_1 \\ \lambda a_2 \\ \vdots \\ \lambda a_n \end{pmatrix}$$

We can put these operations together into what is called a **linear combination**: suppose $v_1, \dots, v_m \in R^n$ and $\lambda_1, \dots, \lambda_m$ are numbers then

$$\lambda_1 v_1 + \dots + \lambda_m v_m$$

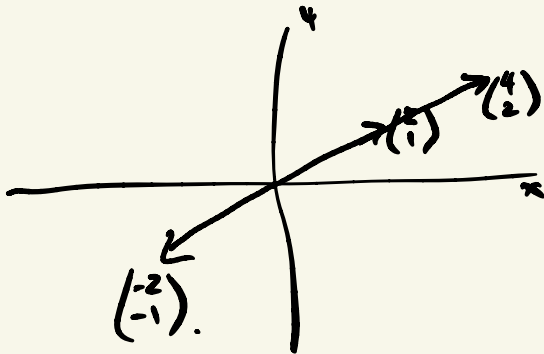
is a linear combination of $v_1, \dots, v_m$.

Ex:

$$2 \begin{pmatrix} 1 \\ 3 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 6 \\ 2 \\ 4 \end{pmatrix}$$

$$(-1) \begin{pmatrix} 1 \\ 3 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} -1 \\ -3 \\ -1 \\ -2 \end{pmatrix}$$

In $\mathbb{R}^2$,



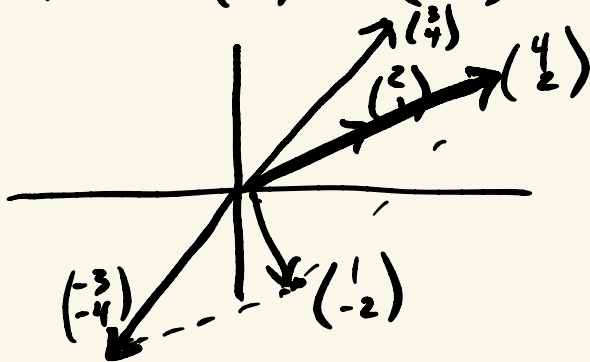
$$2 \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \end{pmatrix}$$

$$(-1) \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ -1 \end{pmatrix}$$

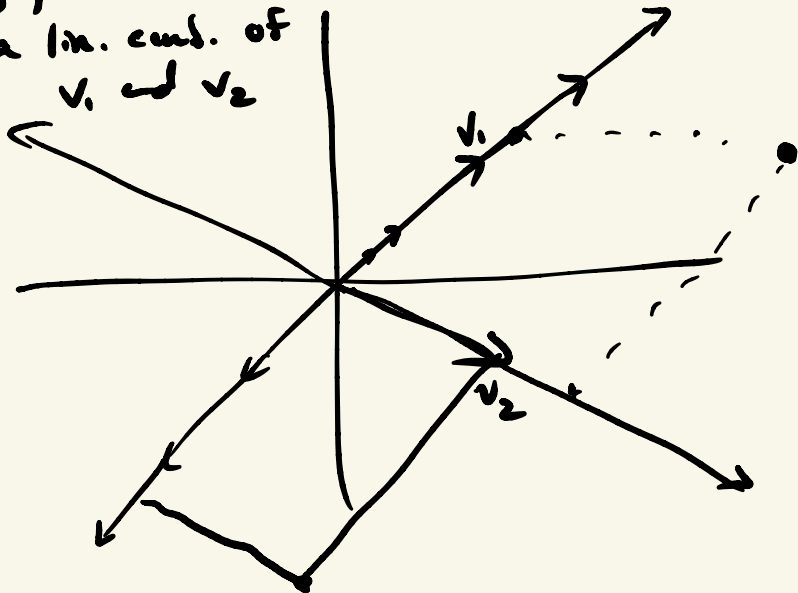
Ex: lin. comb. in $\mathbb{R}^2$

$$\lambda_1 = 2, v_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad \underline{\lambda_2 = -1}, v_2 = \begin{pmatrix} 3 \\ 4 \end{pmatrix}$$

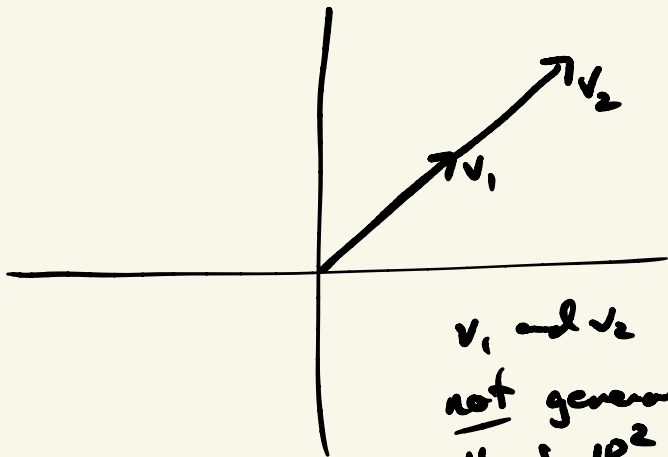
$$2 \begin{pmatrix} 2 \\ 1 \end{pmatrix} + (-1) \begin{pmatrix} 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$



Every point in $\mathbb{R}^2$
is a lin. comb. of
 v_1 and v_2



Ex:



v_1 and v_2 do
not generate
all of $\mathbb{R}^2$ using,
lin. comb.

A fundamental question

- Suppose that $v_1, \dots, v_m$ and b are vectors in R^n . How can we determine if b can be written as a linear combination of $v_1, \dots, v_m$?
- The outcome of our little calculation is that if we treat the matrix

$$A = (v_1 \ v_2 \ \dots \ v_m \ b)$$

i.e. the matrix with columns $v_1, \dots, v_m, b$, as the augmented matrix for a system of linear equations then b is a linear combination of $v_1, \dots, v_m$ iff this system is consistent.

- The **span** of $v_1, \dots, v_m$ is the set of linear combinations of $v_1, \dots, v_m$. It is written $Span\{v_1, \dots, v_m\}$.